ActInf Textbook Group ~ Cohort 6 ~ Session 22 (Chapter 9) 9/16/2024

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 22 (Chapter 9) 9/16/2024 | 1:00:03

okay it's September 14th 24 or
16th my my clock was looking weird um
we're entering chapter 9 for cohort 6
um do either of you want to type or
ask any topics you want to
cover explore for
nine otherwise we can look through the
chapter I was actually about to oh I'm
sorry go good I was actually gonna ask
if we could just like run through the
chapter because this was the first
chapter that I was like oh man we are
jumping into creating this model uh
could we just run through uh one time
into going through like what was the
data and what was the model that we were
actually applying as well as just going
through the loop of
figure I'm sorry I've lost the figure it
doesn't really matter for that last part
yeah got it
okay holding back I think going over the
chapter is the first thing that we'll do
then there's the questions and some
notes and then there's also looking at
software just speaking generally maybe
we can do that in one or two weeks from
now looking at the Jack's branch on pmdp
looking at RX and fur and looking at
other empirical work let's start with
the chapter though okay second half of
the book applying active inference
coming fresh off of chapter 6 recipe
chapter seven and eight giving the
continuous and discreet time now getting

to using the models that are created for
data analysis that's kind of revealing
that in in all the previous chapters 2 3
four um not five but then seven and
eight when talking about building a
generative model it's to produce
synthetic data like to run a simulation
forward here now we're also or
alternatively looking at a setting of
having a certain structure of data and
then seeking to recover latent
parameters instead of specifying the
latent parameters and then generating
the synthetic observation like data um
our general goal is to recover the
parameters of the generative model that
that system of Interest uses to produce
Behavior subjective
model 9.2 metab basian methods this is
really well summarized in figure
91 it's like zooming out a layer from
the figure
4.3 um figure 7.3 type
PDP hash model stream 14.1
um
PDP that we've been specifying and
talking about minimizing the variational
free energy and so on zooming out saying
okay not just the mouse and the Tas now
considering the observations for the
mouse are actions for the researcher and
then conversely the output behaviors of
the mouse are objective observations for
the
ethologist interesting how it's using
the notation of like the subject
centered view because it's it's calling
them oh where whereas you might also

call them actions of the the researcher
and then it's like it gives the the hint
of the suggestion okay yeah you could
make a policy if you wanted to do policy
control
variable um like another Pi sub
scientist and then have that connect to
O but otherwise we're just taking this
as a given so we're just taking that as
the input data for here and so that
subjective model which does the forward
inference which is like um the kind of
perception action Loop that it it's
focused on that's happening with a given
set of observations in an experimental
context and outputting some
Behavior so this kind of showing okay
the data like the input data here's the
the information that we knew on the
which where we put the food in the te-as
and here's the actual sequence of moves
that the mouse
made um brief little statistical
interlude that starts to to bridge back
from this question
of reducing uncertainty of the unknown
parameters of Interest figuring out the
subject's prior beliefs it's like we
know where their iades were that's just
empirical um or we know which things
they bought that's just empirical but
then to do belief modeling cognitive
modeling Beyond just the pure
behaviorist layer then that is reducing
uncertainty about these causal PR M of
Interest so how are those going to be
analyzed um from real finite
data um they go into variational

Applause this is something brought up in
other textbooks um in the SPM
textbook okay 9.3 9.4 are giving like
two different ways more of a classical
statistic ially parametric and then more
of a empirical basian approach to
fitting a first kind of structural layer
on these hidden belief
distributions um variational Appo
centers a uh gaussian in the log
likelihood uh
Thomas uh par discusses this in 57.1 or
point2 fits a um gaussian on log
likelihood so in the actual likelihood
space it doesn't have like a um or it's
a parabola in the log space and then
like it looks different in the
non-logged space but this just fits
uh a a central tendency on a belief
distribution
parametrically that that doesn't require
any
um that doesn't that fits a simple
to uh data parametric empirical base
takes a slightly different approach
which is like to Kickstart the
estimation of the prior from the
empirical overall distribution of the
data or like a previous or related
experiment here's the instructions for
doing the modelbased analysis so collect
data first or maybe the data are already
collected like it's a meta
analysis um so just like the previous
figure the the the data that are
imported here are like you know the
input which which arm was the food on
which color was the strobe light and

then the context of the experiment if if
only to normalize out and say yeah
actually the uh the digit in the in the
minutes column has no correlation with
this but it might be something just
included formulate a PDP model so that's
as we saw here with taking the
subjects kind of firstperson perspective
on taking in observations and how it
does its belief construction here's a
map describing its belief
construction representation and then um
how that information becomes actionable
and then we have just
empirically the information on the
temperature of the room and the the
tmas this information in the list is
summarized in figure 92 so we had data
collection or we're reviewing some data
that were already collected about the
tmas but we have the lab type data then
we have chapter
7.3 um PDP for the teammates
three specify likelihood function
equation
9.2 actions selected depend on the
observations
made we could look more about this but
it kind of gets to the punchline of the
and and it's kind of subtle with the
bold and what and the Bold being
Vector but the the policy that's
selected is the the argument that's
minimizing the variational free energy
over all the
policies so that's like like how
[Music]
um how decision making is going to

be evaluated from the outside as if it's
minimizing free energy as opposed to
other approaches like as opposed to it
maximizing
rewards this tells us how the model
should be used to calculate a likelihood
this typically calls a pmdp solver SPM
mdp vbx it's interesting these are like
super of
referenced
scripts they're in
Matlab there are Matlab engines that can
call Matlab if you have a real license
from other languages so it can be really
called within python um other options
there's octave which is like an open-
Source cousin of mat lab oh well there's
SPM Standalone which is actually mat lab
but it's just like a reduced kind of
single SPM only um download then there's
octave and
patches I don't know how well the
coverage is for patching certain things
of of SPN mat lab into octave there's
Julia SPM which is an attempt to to Port
over to Julia there
py call and Julia call so basically
calling languages within each other um
and the getting these scripts which are
kind of key Central pillars of a lot of
the inference across prior
papers getting these
scripts into um nonproprietary more
easily
callable
um methods and implementations it's
it's
huge four specify prior beliefs about

the parameters terms of expectations and
precisions here setting a uh prior just
to to give an initial condition again
that could be um parametrically
gleaned from data or it could be
empirically set like
PEB inverting a model is is
um using this rearrangement that's
possible with Bas
equation and using kind of algebra and
separation from there to calculate what
we're interested in which is like the
belief the the the the um distributions
of subjective beliefs beliefs of the
mouse in the teen days conditioned upon
all the stuff that we're putting into
model whereas the synthetic generative
direction would be given the um ad hoc
specification of belief distributions
like what if it thought it was three
times more likely then you could
generate realistic trains of
behavior you take those final behavioral
data estimates and it's like here's um
five days of the week we did the
experiment
and um belief whether we calibrated or
didn't calibrate like whether it's just
a um fear score or it's it's calibrated
or it's somehow um normalized like it's
a percentile or something like that and
then this is like a figure just normal
figure from a paper we had five
different condition groups and they're
um belief on this distribution here it
was 7 plus or minus one 5 plus or minus
one 1 plus or minus one negative 1 plus
or minus one and then you can just do a

T Test here's and then you put little
stars statistically different
groups um that's the main outline for
nine I think it'll be awesome to see
fully worked cases like if we can get a
repo that has all of these steps labeled
this way the last pieces 9.6 gives a few
examples 9. three looks at iade in a
continuous and in a discrete control
setting we could look to the paper if we
want to look at that
one table 9.1 reviews many
citations where these kinds of methods
have been
applied that's the
chat so it's interesting because nine is
like it's kind of like a Jimmy Hendrick
thing if six were nine and then six is
like the recipe for building the
generative model and then
nine starting well we still have to
build it as per
six but it's another
recipe but they don't call it a recipe
here
um thought or what should we continue
with um whenever they introduce I think
it was the variational Applause I may be
misremembering they casually brought up
oh yeah you don't really need to
understand this but this is predictive
coding this is just the example we're
using uh if you don't mind could you say
a little bit about if I'm remembering
correctly predictive coding as well as
in your experience like what kind of
models are being used right now in
active inference in general is it all

going to be like just taking uh about
the same model and trying it through
different things are many different
models used whenever you're actually
building these experiments or is it just
really really
yeah good good questions
okay and then let's look at um
H okay 2007 here
2007 all right first let's
see where it came up in
this
okay this is totally pending pending
pending just
um seeing one um just kind of
one take on it look octopus or others
might be able to bring a lot more detail
in this is coming from the parametric
statistic approach so kind of classical
pre-comp
computational that at least pre pre
1900s let's say approaches to fitting
distributions um
whereas once you can do parametric
empirical Bays or like sampling based
methods then there there's this whole
alternative branch of of doing sampling
to fit analytically intractable
distributions but if it's going to be
like well how do you make it more
analytically tractable that that that
might be advantageous or simple might be
um ineffective but also it it it really
does have certain advantages if it if it
if it totally mangles the underlying
structure then again it wasn't adequate
so this isn't to say that it's adequate
for every distribution but if you do a

sampling based
approach then you have to also put a
nice behaving continuous distribution
over that or or choose to not do that
because the sampling gives you a good
Shadow or kind of scatter if you do it
well but it still doesn't have the kind
of
smoothness so it's kind of like if you
want to get to the parametric why not go
directly
from um the data instead whereas now
it's sometimes sampling is thought of as
as an alternative approach but it
doesn't really get to the same end
point this is a incrementally like
Matrix gradient
based method to fit based upon the the
grade
radient of um of a gaan or a parabola or
something downward facing over log
likelihood
and Thomas par mentioned it more on the
stream possibly because it's working
again it's whether it's whether okay
maybe it is a gaussian fitting a g
but then the Gan is locally approximated
with a
quadratic so then that keeps the
approximation of the
gradient um
possible this paper describes
it see if there's anything
here okay
that's just the
preview but it seems like a lot for the
preview
okay so maybe for us to learn more about

but it's
a tractable way to use uh gradient
information geometric gradients
to incrementally optimize statistical
models being proposed existing kind of
alternatively to a sampling based
approach how is it actually being
applied this is a great question
like whenever I come
across code or GitHub I try to bring
into this table everyone everyone else
can too as we build many more we'll
think of better ways to organize it as
as the number of examples gets way
larger than this um I mean there's the
whole RX and fur RX environment's Branch
that's rapidly growing
there's
the and there's a lot of mat lab and SPM
ones that that are not added in here but
there's that large Branch those
ones
mostly call back to those SPM vbx Bays
SPM um mdp check it's like all these
kind of names so there it's like frisen
at all
sheated the functionality of those core
utility files and then papers would use
relatively simple generative models and
then just know that if it was built a
certain way that the um generative and
the data fitting directions would work
so that's the kind of mat
lab um there's there's a long tale of
languages like in the active infer ants
package like
rust go shell just like a bunch of
random languages they're not super

performant but it's enough to get
started
C++ haven't seen any presentations per
se using it but I think some on the on
the more engineering side might
be um of course one of the major uh
questions is like well this is what is
known in open-source research but
anything other than that of course I
don't know on the python side there's
PDP which is the uh main python
package it has a uh characteristic and
lovable SL challenging file structure
that can make it um a little challenging
to get the exact thing you want however
there's a folder Jacks which is the
bringing in the kind of Jacks and
there's more active work on this in the
branches too um so there's there's a lot
of work there kind of to the chapter
nine theme of um fitting from
data
uh there's a lot of other
python
work in um model stream 14
which we which was um shown briefly
earlier ran way said that deep
dreamer he said that
friston said deep dreamer it is
basically active inference so I mean at
least I think ran said this which makes
sense because one of the collaborators
was danger
Hafner so
I think if we kind of just return to the
original what are the applications of
ACM or what what methods are people
using to apply it there's kind of

there's the the threads and the streams
that that can be followed from
here and then there's the parallel
slash is it act in question if you end
up with a pragmatic and epistemic
element to RL or meta RL or PDP type
model or something that can be described
as such I mean has that not reached
active inference from the low road by
another
way what do you
think I kind of wanted to ask you about
that especially at the end of the live
stream he was talking about how there's
different spaces that you could use as
as well as a work that I've never heard
of before by uh Shimon I think he said
uh from the Oxford lab and he was saying
that you can think of the model in terms
of of course the model itself where we
doing a gradient descent so we that's we
going to be able to talk about gradient
flowing in One Direction we can also
talk
about uh the actual concept space of the
model like uh Shimon was talking with
representation spaces I think it was
called and then there's also going to be
the thing that we're learning about
right now which is designing the
environment through the
PDP and then having the model learn from
that so there's going to be some
environment space there's going to be
some belief space that the model uses to
interact with the environment and then
there's also going to be the model
itself which is going to be in the space

that back propagation is really natural
I bring this up because in figure 9.2 uh
like we were looking at earlier whenever
we're actually looking at the model
we're thinking of it in the space of oh
what are we looking to get out of it
which may not be the same space it may
be like uh the PDP like the environment
space that the model is working with
that we're designing is not the native
space that the model is working in which
is going to be the belief space which
reminds me a lot of markof blankets
which maybe that has nothing to do with
anything but I thought that was going to
be an interesting link and I wanted to
get your thoughts on it the movement of
spaces of the changing of spaces As you
move through these squares and figure
9.2
yeah let's return to the question let me
just copy out what this quote was
because I I mean this
is this is this is very interesting to
that ran well for those who have not
seen it it's it's really an informative
stream some of the most information
density on um epistemic value pragmatic
value
connections with active inference and
reinforcement
learning
um okay a lot of people's view of
dreamer versus you know deep active
inference he he mentioned that deep
active inference has been used in
essentially three ways one of them is
when neural networks and other meth and

other um approaches are used that's also
called amortized inference but that's
kind of like deep learning but deep
active inference the other two are like
temporal depth like long Horizon of
planning and then this the third is
hierarchical depth like um and then we
talked about how also those can be
flattened uh deactive inference is that
many people have implemented is that
dreamer so similar to active inference
especially I can't remember if which
version of dreamer and if it does use
Information
Gain term in planning in a planning
objective and if it does then you
basically cannot tell whether there's
any differ between dreamer and deep
active inference um so in a sense even
Carl Kristen has mentioned it sometimes
that dreamer is basically some
variations of an active inference
agent what a truth
drop here was the major approach of the
paper which is the
PDP has an aboutness of uncertainty
subjectively like the a matrix describes
uncertainties associated with the
mapping between observations and hidden
States and then the B describes the
possibly conditional mapping of States
unfolding
um that whole picture though can be seen
as an
mdp because like if you're programming
in PDP you're making a
PDP but for you the PDP is a fully
observable mdp

so then there are certain things that
you can say that are results that only
exist subjectively for mdps like chess
but not for
pdps but then as a model you could say
you can say mdp like things about
pdps and it turns out that the key thing
that the agents Behavior has to follow
along is the Information Gain
but it's really it's it's it really just
shows like how different and then I and
then I said something about Fe and then
and the high road and low road and then
he's like I'm kind of like a low road
guy I don't really have anything to say
on
that which was just like amazing and um
you know very humble and
funny for for how informed and how
capable his presentation research is
that it's
like it wasn't about
FP he didn't really want to go there
yet he was talking about the free energy
functionals with like the
highest degree of
understanding so then that's just like
wow free energy is on the low road
that's the whole thing with energy based
learning variational Auto encoder I mean
millions of people are using evidence
lower
Bound in model
fitting as he as RAM pointed out RL and
meta RL and deep
dreamer go on the low road to
indistinguishable from active
inference so that really puts the low

and the high road into
context
pretty interesting
procedural
method for um I mean this this is like
live stream 42 but here with three L
time layers days hours
minutes and then using a procedural
method to fill this in as opposed to
like doing this and then that one and
then just this and then emitting just
from here or something so I mean there's
so many low
road
methods every method is on the low
road but but none of it is in
competition with the high
road like before we started Z Zach and I
were talking about a little bit about
the reward learning also and and um kind
of like okay want to build the
skyscraper
taller it's just reward function it's
taller and then in contrast is sort of
like um I want to reduce my uncertainty
about looking from the 100th floor and
then maybe you build um a 75 story and
jump or you build a 200 and go to the
100th floor or something like that but
then if there's like a um an equilibrium
that can be reached when that preference
is
achieved it's just so interesting to see
like how those can yield this they could
yield similar or
different computational policies
inferred to get there like there's
situations

where
um the the the RL and the ACM agent
would have indistinguishable
Behavior but there's situations where
they might have different
Behavior
but at the that would all be situational
on literally just specifying okay what
is the setting what are the
data it maybe maybe certain generic
things could be said like well for these
kinds of environments this kind of
outperformance would be expected or this
kind of asymmetry would be expected some
more generic statements about kinds of
of environments where there' be
different kinds of competitiveness but I
mean it still is just a basically
empirical question
do you think biology is going to help us
solve that empirical question because I
remember
whenever it was inside of the book that
introduced the uh system one and system
two uh Sy fast and slow that's what it
was and oh I'm sorry
inside of that they were talking
about the example of I believe it was
Falcons uh that were in
California and so whenever they were
discussing it they asked what was the
value that you would like to apply
towards saving this m and then what they
did was either they intro introduced
increasing the space or increasing the
number of Falcons that you would save
and they asked how much would you uh
would you be willing to spend in order

to donate to this cause and they started
to vary the amount of difference that it
make and they really show that like even
our own belief systems are unintuitive
to us of how we actually value things
and how much we value that especially at
different scales is not something that
we would expect from ourselves do you
think that we would be able to find an
empirically good way to solve this
strictly from how we function
biologically or how any mammal might
function inside of this
Frank or
Axel sorry I just got in here I've been
setting up so I I didn't catch it oh
it's all
good z i it's a great question and
and is slow is active inference related
to being a theory of inquiry along the
lines of the pragmatists what can be
empirically and theoretically said about
empirical situations like laboratory
behaviorism and also Theory producing
situations considered
empirically I don't know though if even
a very strong account there would be equ
to the resolution of uncertainty or the
account for the like the last
mile so you as to whether this RL or
this ACM model will do better on
explaining this bacteria data
set that that's like to know that would
be like to know it's like the halting
problem I mean if you could just it's
like it would be equivalent to knowing
how a data would fit a model before
doing it you could use hero

sticks but but and those hero sticks may
be 100% right in certain
settings but then but that still
wouldn't be like actually
knowing
it but then you could have certain be
like okay
for linear models or for Transformer
models like adding another layer some
times takes more training time but it
never does
worse then you could say certain things
about two hypothetical architectures
being applied to a data set or to models
being applied but it's kind of like the
Basi and model reduction question or
like structure
learning you could have a procedure for
adding and pruning down latent accounts
and and all these kinds of
things but like if you already knew
which model to apply before
then you might be coincidentally right
with your prior but you wouldn't have
been
um right except by
prior okay I see what you're saying now
I think I was analizing too much because
I was thinking of 9.2 and I was thinking
that like oh the ideal situation would
be to where you design a system it's
like oh it works like I work so that's
where I was coming
from interesting I mean because it could
be like it works like I think it works
would be like okay this is a um
metronome and just changing temperature
didn't matter turning the dial totally

explained the frequency of this so it's
like there's there's situations where
this box is well or this could be just
an and gate or could be just some other
discret or or designed system and then
the then on the other side that could do
would be like this is a person talking
to another
person and then but that so that is that
which one is like it works like I
work like seeing the world work as
expected or seeing the world work in a
way that is open and surprising
and I think these are become such
interesting questions as the
technical
um as what the our our active
computational Niche is doing around
us gets more active than like what is it
that we're really
doing but like o opening eye of course
I'm ionically doesn't have I don't think
they release the technical
information um it says you can read more
of technical research
post log
linear but a kind of subtle log
compute better
evals on domain
expertise better coding it's available
in cursor 01 mini is doesn't cost any
extra
they hide the chains of
thought but it's kind of like but you
don't even know whether what what has
happened but it's
doing I mean so then if it's doing
um kind of something between working

memory
transient um Chain of Thought with
pruning planning in thought plan
dreaming to plan planning to
dream maybe using a uh energy based
method or an
elbow and then
like also with with kind of a niche
modification maybe chains of thoughts
can read and write to each other or
something like that there's just so many
architectures people have proposed
yeah
Axel yeah I just have a quick question
here in the end I'm I'm sorry I missed
pretty much most of this I was on in
transport but I have a bit of a more
technical question um I'm interested in
the differences between RL and active
inference and right now I'm going
through some of these Pi
mdp um collab notebooks and one of the
first ones is a tutorial where you're
setting up um these different matrices
for for example um like a is your Matrix
that uh Maps the states to observations
and B is your belief Matrix about the
beliefs of what will happen if you uh
carry out a certain action and so on and
so forth and yeah I think if you open it
maybe the solutions will already be
there you can see what I'm talking
about um but in this example of course
it's a simple one uh it's unambiguous so
you start out with all of the um
matrices that you need to solve the
problem or at least that's how it seems
to

me and I'm kind of interested in how
little you need to specify when I first
got interested in active inference I was
kind of uh attracted by the idea that
you might not have to specify anything
at all you would have an agent that
learns how to carry out uh useful
actions not by having a reward function
but just by doing things that makes it
able to control its environment better
which in the end will be things that uh
perhaps are useful and I'm I think my
question is just how little do you need
to specify is it possible to specify not
even the
agents uh
the agent's understanding of when it
sees that it is in square one it also
will think that it should expect itself
to be in square one like what is the
least thing that I can specify in this
setup and still get an agent that learns
all of these things
itself yeah great
question
um first part on RL and ACM I think
check out this model stream 14 that that
we discussed a little bit because it was
some of the most distilled and um
experienced comparisons but then this
question of like what is needed to be
specified so um you're totally right
that that um this is kind of like it's
like a Canon inference Canon or reverse
Cannon set up on
Rails with with a certain like it's kind
of aimed on a certain axis already
because it's like well how does it know

it's in a teamas how does it know the
the semitics of how the Q at the bottom
relates to um whether it's on the left
or the right you know how does it know
how to walk so it's like but that's
where it becomes really important to
clarify what are questions about the
world what are questions about
generative models and then where are we
developing generative models to address
questions about the world and then those
can go back and forth but it's like if
you wanted to study well how does the
semiosis become associated between the Q
and the teas you could study that with a
degree of Freedom around that relation
or you could just say we're assuming
it's a perfectly trained rat that
already 100% understands like the left
and the right cues but those are
generative models that are being
constructed for different cases so
depending on what is is wanting to be
um get gotten intuition for like in a
toy model or what kinds of research
questions are being addressed from it's
like it be like downloading a genome and
just like model it it's like but I mean
to what end the 3D shape or like you
know the regulatory Network it's like
well all of them it's like yeah but then
that's just the whole project of the
field or something that's that's not a
specific research question to ask about
so yes definitely and then when it comes
down to in into this level it's like
yeah so then but then what what about
adding a

fourth grid location or what how would
it come to you know then then it has to
change it so th those are some of the
rate limiting steps in the in
implementing things like structure
learning in the models today which is
like the dimensionality has all these
repercussions or how does it how would
it look how would it come to know that
it could add a new affordance so
can it be something very simple with a
procedural approach to like changing the
dimensionality and the number of slices
ABCDE and then could that like a
procedure over
ABCDE have the ability to
learn in this or that way it's an open
question to and then okay you could set
it up but but then if you set it up in a
in a computational environment that only
has like multi-arm Bandit slot
machines I mean it's not going to code
in
Python
so then that gets into the niche
specification either the computational
niche for for training and learning and
and inference or more of the
robotics route with the real world
Niche so uh yeah I I get what you're
saying in terms of uh it's interesting
how you might want to procedurally
change your structure when you've
already uh established it but then let
me ask what if you have established the
exact structure you you live in this
simple grid world you can establish all
of that and perhaps there are certain

things you can do in the grid World
maybe you have to get keys in order to
open doors in order to go somewhere but
there's no final reward in the goal
position you want to go to do you think
an active inference agent because it's
curious and because it wants to learn to
understand its world will figure out how
to do the actions which will end up
getting it to be able to predict what's
going to or like get the furthest in the
world just because it
uh has that incentive
perhaps okay great question um so one we
we never really brought a model together
but we explored it but it was 2021 so
before like
PDP and and rser and all this but it was
like net hack it's like this text based
Explorer that's famously convoluted but
even just considering like yeah how do
you do items as affordances but then do
every turn are you going to do like this
Mega Deepa in on like every possible
spell that you could cast in every
direction or how are we even going to
learn to
know possibly things that even people
haven't written down
like how two spells interact with the
dragon only on the 13th level or
something it's like so this could be a
very interesting setting again it's it's
it's um hard in general but the the
reinforcement learning a reward approach
let's just say that it's like um you
know kind of this surrealist uh dream
grid World W with the keys and the doors

and all this other stuff I mean there's
physiological sort of grounded layer
which like what makes more sense saying
I'm rewarded by being at physiology or I
expect to be at physiology and that's
just where I satisfy so let's just say
physiology can be equally well explained
by actm or by a reward term
um then it's like well where do you go
then with your
reward and then it's like if you set
your reward arbitrary function on
curiosity learning epistemic value then
you're doing function approximation for
epistemic Value so then you've just
created a sort
of epistemic approximator RL agent
there's nothing wrong with that it might
be totally fine but um it it's it's the
low road side of a question that could
be approached from a high road which
would just be the epistemic value
expected information game things that
might be like important for the net hack
challenge um but then how would you come
to tune and adjust the epistemic and
pragmatic balance on the
fly
so it it isn't that you know like like
again think about the RL and the actm
it's like let's just say there there
there's a basketball hoop that's 20 feet
tall and someone's like I prefer myself
to be dunking or the reinforcement
learning is just like I want to redu I
want to I want to jump higher and then
I'll be able to dunk but it's like that
doesn't mean it's possible to even do it

but the reinforcement learner would
train to jump up to 10 feet and then
they would just stall and they just
would be like you know I think I could
do better or I'm at the best score I've
ever gotten or something the ACT imp
would just have a a would reach a
plateau of
surprise like I'm why am I three bits
surprised about my jumping when I expect
to be doing a 20 foot
dunk but it's like but this but if it's
an irresolvable situation it's then it's
it's been set up that it can't be done
whereas if it was a t if it was an
achievable
situation then both of those would would
have a possibility to be able to
dunk but at least leaving the door open
for the epistemic and having an A Part A
stabilized uh open
door that isn't just pragmatic
because if you operationalize the
epistemic value into pragmatic value
okay train the reinforcement learning on
learning then we get back to these
questions right that we're having at the
very beginning like well then it would
learn at cost to other
variables oh no no well then we'll learn
to balance that okay but then then you
you can just have this infinite like
regress of of reinforcement learning but
it still is in pragmatic Val value from
the high road
side right yeah I think I'm just having
a hard time understanding exactly there
was this paper once about uh how you

could teach agents to learn to ride a
bike just by trying to maximize
empowerment so trying to make it so that
these agents would uh not have a reward
for having the bike upright but would
have a reward for being able to do more
things in the world and since they had
the bike upright they could drive
further and go more places and because
of that uh they could do more things and
I'm wondering if the same kind of thing
happens in active inference where if you
had a agent in a in a grid with a place
where a key comes out if it would just
go up and see that a key comes every 10
seconds and think that that's nice and
now it's learned that and there's no
more surprise or if it would somehow be
incentivized to say oh but what happens
if I take this key and go down into the
bottom of the maze where there's this
door even though it didn't know anything
about that that that would open anything
or yeah this um Ali was there for this
stream with Tom
ringstrom it's it's a it's a totally
viable
path um it's interesting also about the
Bellman equations which come up in the
mdp
solutions so treating an a PDP like an
mdp which is say using Bellman
optimality PDP uh mdp Style on the agent
pmdp um there might be situations where
where expected free energy and optimal
and empowerment are are
like correlated or there there might be
um designable situations where they're

different
enable I think one important note and
it's kind of the separation between the
equation 2.5 and 2.6 is like 2.5
variational free energy is not
prospective this is just like a real
time existence
check so this is more like a a a through
the flow of the
present once you get into planning over
expected future futes like there's
expected free energy then there's free
energy of the expected future
fef there then there's other approaches
that have been proposed so I think
there's a lot of constructions that
highlight different parts of the
future to say nothing of the of the
procedural ways that you could like
prune through
Futures but there might be theistic of
um max maximum occupancy
or um maximal
empowerments or maximal um uh time
discounted reward for some agent or time
discounted learning or like
um the amount of this or that within a
certain time
frame so I think there's just there's a
lot more ways to model the
future thank you so much yeah I'll
definitely check out this
paper yeah this it's a great connection
though and I think it's a similar there
might be other intrinsic motivators
like like you know wood learning wood
wanting to learn a space equate to
occupying it through a Time equate to

empowerment and movement in that
space again you could imagine a space
where in an agent where like that those
are all
IND
distinguish but then they could have
some other space where just like oh but
if you you know the cells change at
different rates so if you are equally
occupying the
space then you're not equally learning
it but it's like but then that that's
just getting into these more generic
like
comparators but I mean these are great
these are great questions and and
um that these are all weaving together
in in a chapter nesque
way as we think about generating data
from ACM models fitting acin models to
data I
mean if if we
can if we can
um share and
borrow INF structurally with dreamer and
all of these
mainstream machine learning models and
contextualize it from a high road and
add New Paths on the low
road just just a few memes
away okay thank you all see you you uh
next
time you so much bye bye bye